

IT Asset Management - Query Processing

DESCRIPTION

Fall 2023

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Class and Team Information

CSCI 3410-Section DA, Group 1

Team members:

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2. Lindy Ankorn
3. Brent Gibbins
4. Dawson Jones-Fischer
5. Evan Nowicki

Problem Description

Write and execute queries against the **database you will create** using the following conceptual and logical designs.

Populate your tables with a small set of appropriate and relevant data and run the queries against them. In about a week before the project deadline, I will give you my **sample dataset**, which you must use to produce

the results to be submitted. Therefore, it is imperative that you develop and test your queries thoroughly with your own data set before you receive mine.

The **columns of the results** must match the sample query results I have shown in each query section.

If the **requested number of output rows** is too many to be screen captured, export the result to a CSV file first, then extract (i.e., screen shot) and include the requested number of rows in this document.

Conceptual Design and Logical Design

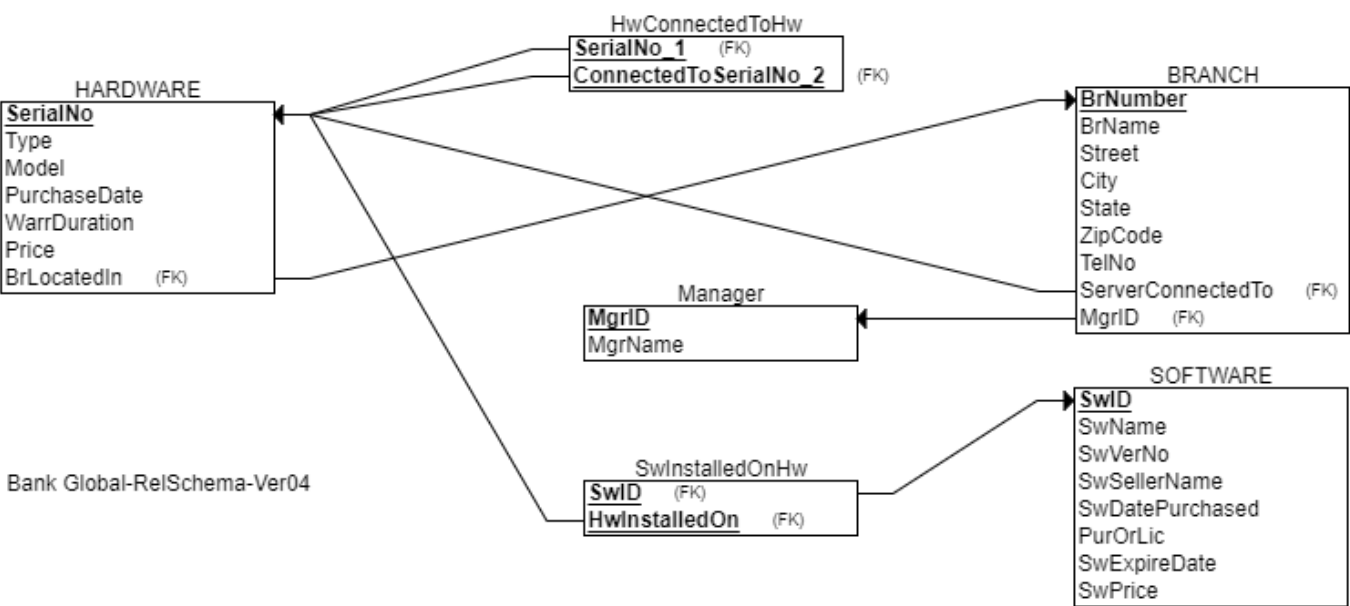
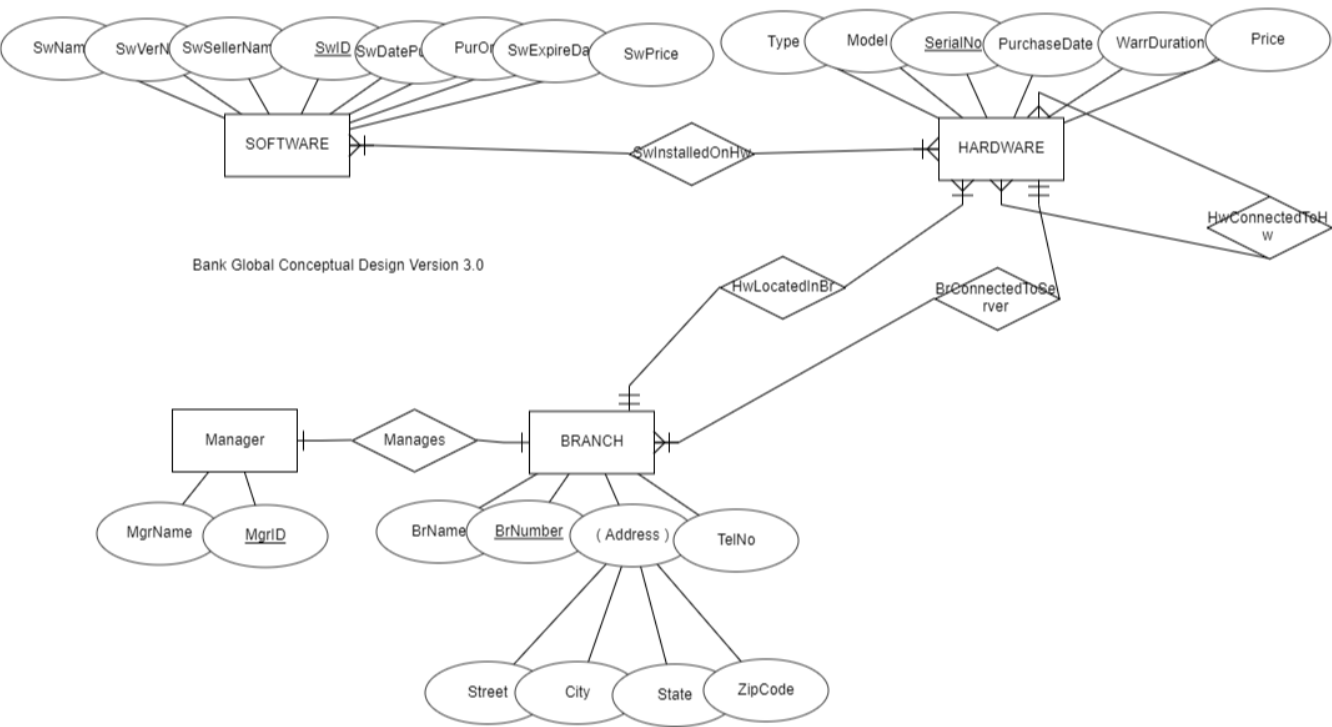


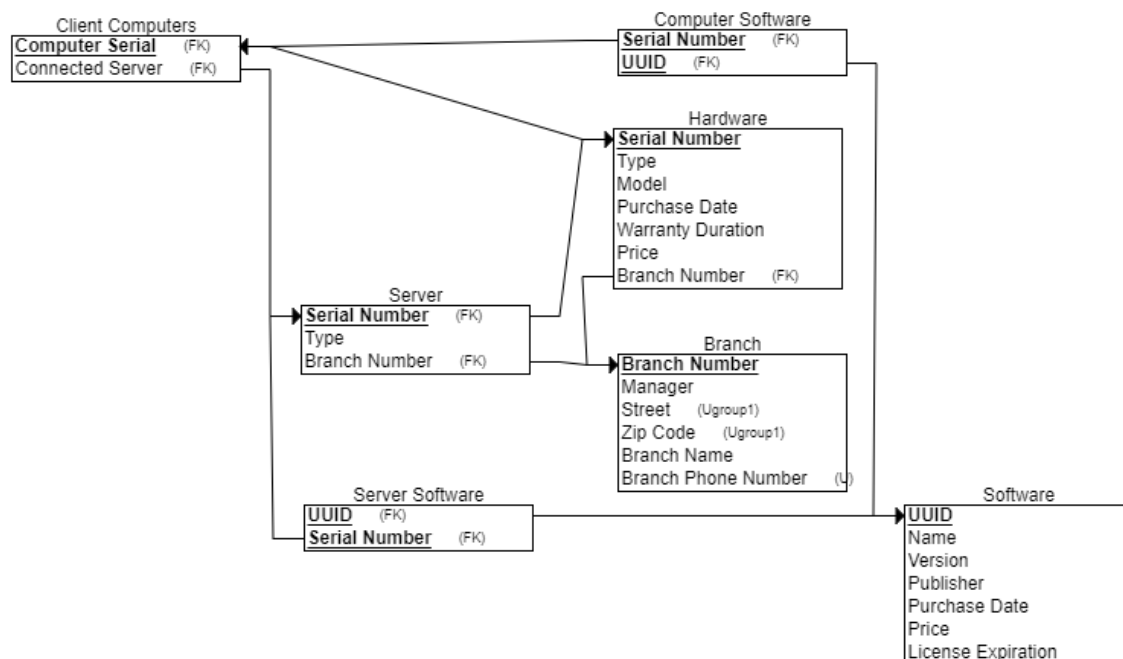
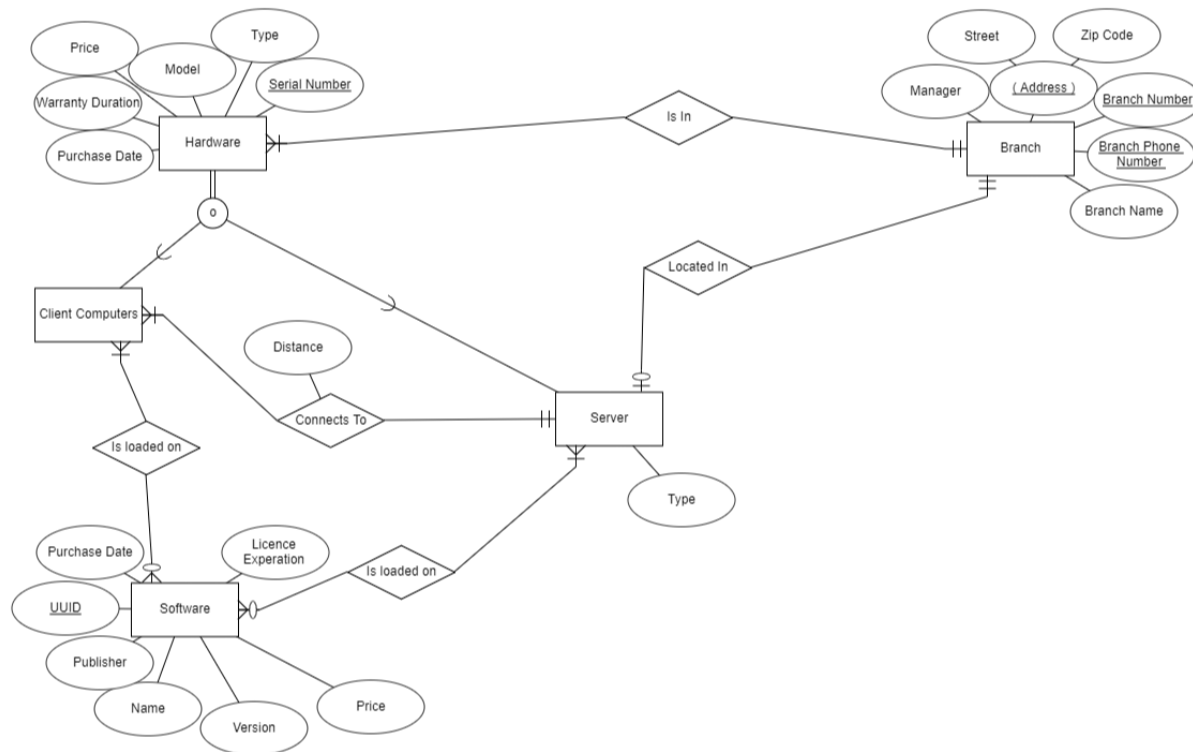
Table Name BRANCH			Table Name HARDWARE			Table Name HwConnectedToHw		
PRIMARY KEY (U) GROUPS ...			PRIMARY KEY (U) GROUPS ...			PRIMARY KEY (U) GROUPS ...		
RECURSIVE KEY REORDER ...			RECURSIVE KEY REORDER ...			RECURSIVE KEY REORDER ...		
UNIQUE ... OPTIONAL ...			UNIQUE ... OPTIONAL ...			UNIQUE ... OPTIONAL ...		
Columns			Columns			Columns		
ADD REMOVE ...			ADD REMOVE ...			ADD REMOVE ...		
Name	Data Type	Data Type Size	Name	Data Type	Data Type Size	Name	Data Type	Data Type Size
<u>BrNumber</u>	INT		<u>SerialNo</u>	VARCHAR(n)	50	<u>SerialNo_1</u>	(Foreign Key)	
BrName	VARCHAR(n)	50	Type	VARCHAR(n)	25	<u>ConnectedToSerialNo_2</u>	(Foreign Key)	
Street	VARCHAR(n)	50	Model	INT				
City	VARCHAR(n)	25	PurchaseDate	DATE				
State	VARCHAR(n)	25	WarrDuration	INT				
ZipCode	VARCHAR(n)	10	Price	NUMERIC(p s)	(10, 2)			
TelNo	VARCHAR(n)	25	BrLocatedIn		(Foreign Key)			
ServerConnectedTo		(Foreign Key)						

Table Name Manager PRIMARY KEY (U) GROUPS ... RECURSIVE KEY REORDER ... UNIQUE ... OPTIONAL ... Columns ADD REMOVE ... <table> <tr> <th>Name</th><th>Data Type</th><th>Data Type Size</th></tr> <tr> <td><u>MgrID</u></td><td>INT</td><td></td></tr> <tr> <td>MgrName</td><td>VARCHAR(n)</td><td>25</td></tr> </table>	Name	Data Type	Data Type Size	<u>MgrID</u>	INT		MgrName	VARCHAR(n)	25	Table Name SOFTWARE PRIMARY KEY (U) GROUPS ... RECURSIVE KEY REORDER ... UNIQUE ... OPTIONAL ... Columns ADD REMOVE ... <table> <tr> <th>Name</th><th>Data Type</th><th>Data Type Size</th></tr> <tr> <td><u>SwID</u></td><td>VARCHAR(n)</td><td>50</td></tr> <tr> <td>SwName</td><td>VARCHAR(n)</td><td>25</td></tr> <tr> <td>SwVerNo</td><td>INT</td><td></td></tr> <tr> <td>SwSellerName</td><td>VARCHAR(n)</td><td>25</td></tr> <tr> <td>SwDatePurchase</td><td>DATE</td><td></td></tr> <tr> <td>PurOrLic</td><td>CHAR(n)</td><td>1</td></tr> <tr> <td>SwExpireDate</td><td>DATE</td><td></td></tr> <tr> <td>SwPrice</td><td>NUMERIC(p s)</td><td>(10,2)</td></tr> </table>	Name	Data Type	Data Type Size	<u>SwID</u>	VARCHAR(n)	50	SwName	VARCHAR(n)	25	SwVerNo	INT		SwSellerName	VARCHAR(n)	25	SwDatePurchase	DATE		PurOrLic	CHAR(n)	1	SwExpireDate	DATE		SwPrice	NUMERIC(p s)	(10,2)	Table Name SwInstalledOnHw PRIMARY KEY (U) GROUPS ... RECURSIVE KEY REORDER ... UNIQUE ... OPTIONAL ... Columns ADD REMOVE ... <table> <tr> <th>Name</th><th>Data Type</th><th>Data Type Size</th></tr> <tr> <td><u>SwID</u></td><td></td><td>(Foreign Key)</td></tr> <tr> <td><u>HwInstalledOn</u></td><td></td><td>(Foreign Key)</td></tr> </table>	Name	Data Type	Data Type Size	<u>SwID</u>		(Foreign Key)	<u>HwInstalledOn</u>		(Foreign Key)
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Designs, Queries and Sample Results

Original ER Diagram and Relational Schemas (25 points)

[Insert, below, the images of your conceptual design (i.e., ER diagram) and logical design diagram (i.e., relational schema diagram), which you developed, initially, before receiving mine.]



List of Tables, Corresponding Columns and Column Types (75 points)

[Example:

Table: **branch**

Columns:

BrName	varchar(50)
<u>BrNumber</u>	int PK
Street	varchar(50)
City	varchar(25)
State	varchar(25)
ZipCode	varchar(10)
TelNo	varchar(25)
ServerConnectedTo	varchar(50)
MgrID	int

]

[Similar to the above example image, insert the images of your tables below, in the ascending order of table names]

Table: **branch**

Columns:

<u>BrNumber</u>	int PK
BrName	varchar(50)
TelNo	varchar(25)
Street	varchar(25)
City	varchar(25)
State	varchar(25)
ZipCode	varchar(10)
ServerConnectedTo	varchar(50)
MgrID	int

Table: **hardware**

Columns:

<u>SerialNo</u>	varchar(50) PK
Type	varchar(25)
Model	int
PurchaseDate	date
WarrDuration	int
Price	decimal(10,2)
BrLocation	int

Table: **hwconnectedtohw**

Columns:

<u>SerialNo_1</u>	varchar(50) PK
<u>ConnectedToSerialNo_2</u>	varchar(50) PK

Table: **manager**

Columns:

MgrName	varchar(25)
<u>MqrID</u>	int PK

Table: **software**

Columns:

SwName	varchar(25)
SwVerNo	int
SwSellerName	varchar(25)
SwDatePurchased	date
PurOrLic	char(1)
SwExpireDate	date
SwPrice	decimal(10,2)
<u>SwID</u>	varchar(50) PK

Table: **swinstalledonhw**

Columns:

<u>SwID</u>	varchar(50) PK
<u>SerialNo</u>	varchar(50) PK

Query 1 – Servers

Query in English: List the types and model numbers of all the servers. (ALL ROWS)

Sample Query Result:

HwType	model
server	18
server	29
server	19
server	23

Query in SQL:

[Show the query statements]

```
1  SELECT Type as HwType, Model as model
2  FROM hardware
3  WHERE Type = 'server';
```

Result:

[Show the result of the query]

	HwType	model
▶	server	18
	server	29
	server	19
	server	23
	server	11
	server	11
	server	24
	server	14
	server	17
	server	25

Query 2 – Hardware Systems

Query in English: List the branch numbers and zip codes of the branches, which house all the hardware systems (1st 50 ROWS)

Sample Query Result:

BranchNumber	BranchZipCode	HousedHardwareType
100	82196	Aeneand
100	82196	dapibusd
100	82196	nunc.d

Query in SQL:

[Show the query statements]

```
1 • SELECT b.BrNumber as BranchNumber, b.ZipCode as BranchZipCode, h.Type as HousedHardwareType
2 FROM branch b, hardware h
3 WHERE h.BrLocation = b.BrNumber;
```

Result:

[Show the result of the query]

A	B	C
BranchNumber	BranchZipCode	HousedHardwareType
100	82196	Aeneancl
100	82196	dapibuscl
100	82196	nunc.cl
100	82196	EntServerType
100	82196	server
101	82196	risus.cl
101	82196	risus.cl
102	82196	Nuncl
102	82196	portacl
103	82196	ipsumpr1
103	82196	nasceturcl
103	82196	luctuspr2
103	82196	portacl
105	65261	server
106	65261	eucl
107	65261	blanditcl
107	65261	risuscl
107	65261	loremcl
108	78451	server
108	78451	ipsumcl
109	78451	necatrccl
109	78451	ipsumcl
109	78451	tortorpr1
109	78451	ultricespr2
110	78451	bloomcl
111	78451	silvercl
112	78451	super
113	66398	server
113	66398	ametcl
113	66398	portacl
114	28667	server
116	77034	goldcl
116	77034	etcl
117	77034	ornarecl
117	77034	Suspendisseclcl
117	77034	silvercl
118	77034	trunccl
119	77034	sericl
120	77034	trascl
121	77034	molestiepr3
121	77034	enimpr1
121	77034	ipsumpr2cl
121	77034	road
121	77034	risus.
122	90759	server
123	90759	jone
124	90759	jinu
126	72206	et
126	72206	server
127	72206	ipsum.

Query 3 – Branches and Housed Servers

Query in English: List the branch numbers and zip codes of the branches, which house the servers (ALL ROWS)

Sample Query Result:

BrNumber	ZipCode	HwType	ServerModel
114	28667	server	18
113	66398	server	29
105	65261	server	19

Query in SQL:

[Show the query statements]

```
1 • SELECT branch.BrNumber, branch.ZipCode, hardware.Type AS HwType, hardware.Model AS ServerModel
2 FROM branch
3 JOIN hardware ON hardware.BrLocation = branch.BrNumber
4 WHERE hardware.Type = 'server';
```

Result:

[Show the result of the query]

	BrNumber	ZipCode	HwType	ServerModel
▶	114	28667	server	18
	113	66398	server	29
	105	65261	server	19
	108	78451	server	23
	122	90759	server	11
	140	71853	server	11
	126	72206	server	14
	133	76041	server	17
	100	82196	server	25

Query 4 – Asset Prices

Query in English: What is the count and total price of the hardware and software assets in the bank? (ONE ROW)

Sample Query Result:

AssetCount	AssetsTotalPrice
233	\$1,811,144

Query in SQL:

[Show the query statements]

```
1 • SELECT (SELECT COUNT(SerialNo) FROM hardware) + (SELECT COUNT(SwID) FROM software) as AssetCount,
2 CONCAT('$', FORMAT((SELECT SUM(Price) FROM hardware) + (SELECT SUM(SwPrice) FROM software), 2)) as AssetsTotalPrice;
```

Result:

[Show the result of the query]

	AssetCount	AssetsTotalPrice
▶	233	\$1,811,144.00

Query 5 – Expired Software

Query in English:

Create a procedure called ExpiredSoftware to show the software applications, which have expired GraceDays before a TargetDate. TargetDate and GraceDays are the input parameters of the procedure.

Show the results of the following two procedure calls:

CALL ExpiredSoftware("2020-12-01", 90); -- ALL THE ROWS

CALL ExpiredSoftware("2023-09-05", 60); -- 1ST 20 ROWS

Sample Query Result:

SwName	SwPrice	TargetDate	SwExpireDate	GraceDays	DaysPassedExpiration
Software Name 15	5431.00	2020-12-01	2020-01-10	90	326
Software Name 37	3021.00	2020-12-01	2020-04-15	90	230
Software Name 38	11294.00	2020-12-01	2020-04-21	90	224

SwName	SwPrice	TargetDate	SwExpireDate	GraceDays	DaysPassedExpiration
Software Name 1	2693.00	2023-09-05	2022-02-05	60	577
Software Name 5	211.00	2023-09-05	2022-03-07	60	547
Software Name 6	12605.00	2023-09-05	2022-07-08	60	424

Query in SQL:

[Show the query statements]

```

1  DELIMITER //
2
3  • CREATE PROCEDURE ExpiredSoftware(IN TargetDate DATE, IN GraceDays INT)
4  • BEGIN
5      SELECT SwName, SwPrice,
6              TargetDate,
7              SwExpireDate,
8              GraceDays,
9              DATEDIFF(TargetDate, SwExpireDate) AS DaysPassedExpiration
10     FROM software
11     WHERE SwExpireDate <= DATE_SUB(TargetDate, INTERVAL GraceDays DAY);
12 • END //
13
14 DELIMITER ;
15
16 • CALL ExpiredSoftware("2020-12-01", 90);
17
18 • CALL ExpiredSoftware("2023-09-05", 60);
19

```

Result:

[Show the result of the query]

	SwName	SwPrice	TargetDate	SwExpireDate	GraceDays	DaysPassedExpiration
►	Software Name 15	5431.00	2020-12-01	2020-01-10	90	326
	Software Name 37	3021.00	2020-12-01	2020-04-15	90	230
	Software Name 38	11294.00	2020-12-01	2020-04-21	90	224

	SwName	SwPrice	TargetDate	SwExpireDate	GraceDays	DaysPassedExpiration
►	Software Name 1	2693.00	2023-09-05	2022-02-05	60	577
	Software Name 5	211.00	2023-09-05	2022-03-07	60	547
	Software Name 6	12605.00	2023-09-05	2022-07-08	60	424
	Software Name 8	7860.00	2023-09-05	2022-03-26	60	528
	Software Name 9	6876.00	2023-09-05	2021-12-31	60	613
	Software Name 10	4891.00	2023-09-05	2022-11-19	60	290
	Software Name 11	4299.00	2023-09-05	2022-01-29	60	584
	Software Name 13	13851.00	2023-09-05	2020-09-05	60	1095
	Software Name 15	5431.00	2023-09-05	2020-01-10	60	1334
	Software Name 16	8232.00	2023-09-05	2021-04-05	60	883
	Software Name 18	9424.00	2023-09-05	2022-07-16	60	416
	Software Name 22	3401.00	2023-09-05	2022-07-28	60	404
	Software Name 23	925.00	2023-09-05	2022-02-11	60	571
	Software Name 25	4423.00	2023-09-05	2022-05-09	60	484
	Software Name 26	8357.00	2023-09-05	2021-12-31	60	613
	Software Name 29	14369.00	2023-09-05	2023-05-13	60	115
	Software Name 31	9964.00	2023-09-05	2022-01-07	60	606
	Software Name 32	6475.00	2023-09-05	2021-04-14	60	874
	Software Name 37	3021.00	2023-09-05	2020-04-15	60	1238
	Software Name 38	11294.00	2023-09-05	2020-04-21	60	1232

Query 6 – Impacted branches by the expired software

Query in English:

Create a procedure called ImpactedBranches to list the branch numbers, which are impacted by the expiring software applications. The branches will be impacted, if an expiring software is installed on a hardware system, which is either located in the branch or the branch is connected to it. In the latter case, the hardware system must be a server to which the branch is connected. TargetDate is the input parameter of the procedure. A software application is expired, if its expiration date is before the TargetDate.

Your query should sort the result in ascending order of BrNumbers.

Show the result of the following call:

CALL ImpactedBranches ("2023-01-01"); -- (1ST 20 ROWS)

Sample Query Result:

BrNumber	HwSerialNo	SwName	TargetDate	SwExpireDate
100	AA88FEDB-C5EC-9476-C895-3E62A2A7814A	Software Name 1	2023-01-01	2022-02-05
100	EntServer	Software Name 56	2023-01-01	2022-07-09
100	SEA2C586-88ED-A0A3-0C13-EECA13E394AB	Software Name 54	2023-01-01	2022-03-11
101	SEE41781-8DDC-243F-B0F2-9F668BB5D279	Software Name 55	2023-01-01	2022-09-17
101	SEA2C586-88ED-A0A3-0C13-EECA13E394AB	Software Name 54	2023-01-01	2022-03-11
102	SEA2C586-88ED-A0A3-0C13-EECA13E394AB	Software Name 54	2023-01-01	2022-03-11
103	SEA2C586-88ED-A0A3-0C13-EECA13E394AB	Software Name 54	2023-01-01	2022-03-11
103	555538D6-043D-3C05-6CB0-40A554D16700	Software Name 56	2023-01-01	2022-07-09

Query in SQL:

[Show the query statements]

```

1 • DROP PROCEDURE IF EXISTS ImpactedBranches;
2
3 DELIMITER //
4
5 • CREATE PROCEDURE ImpactedBranches(IN TargetDate DATE)
6 BEGIN
7     SELECT DISTINCT branch.BrNumber, hardware.SerialNo AS HwSerialNo, software.SwName, TargetDate AS TargetDate, software.SwExpireDate
8     FROM branch
9     JOIN hardware ON branch.BrNumber = hardware.BrLocation
10    JOIN swinstalledonhw ON hardware.SerialNo = swinstalledonhw.SerialNo
11    JOIN software ON software.SwID = swinstalledonhw.SwID
12    LEFT JOIN hwconnectedtohw ON branch.ServerConnectedTo = hwconnectedtohw.ConnectedToSerialNo_2
13    WHERE (hardware.SerialNo IS NOT NULL OR hwconnectedtohw.ConnectedToSerialNo_2 IS NOT NULL)
14    AND software.SwExpireDate < TargetDate
15    ORDER BY branch.BrNumber ASC;
16 END //
17
18 DELIMITER ;
19
20 • CALL ImpactedBranches("2023-01-01");

```

Result:

[Show the result of the query]

	BrNumber	HwSerialNo	SwName	TargetDate	SwExpireDate
▶	100	AA88FEDB-C5EC-9476-C895-3E62A2A7814A	Software Name 1	2023-01-01	2022-02-05
	100	EntServer	Software Name 56	2023-01-01	2022-07-09
	100	SEA2C586-88ED-A0A3-0C13-EECA13E394AB	Software Name 54	2023-01-01	2022-03-11
	100	SEE41781-8DDC-243F-B0F2-9F668BB5D279	Software Name 55	2023-01-01	2022-09-17
	100	2833D1B3-43B7-A852-71AE-16EDC15EDBCB	Software Name 15	2023-01-01	2020-01-10
	100	2A2320A7-244C-D8AE-DF54-5BF116D36B14	Software Name 16	2023-01-01	2021-04-05
	100	SEEE28D6-943D-36CF-66B9-40A5E4DA679B	Software Name 56	2023-01-01	2022-07-09
	100	733D835C-E821-BD5E-CBC1-D9D16A8164B7	Software Name 52	2023-01-01	2022-03-20
	100	D322F781-5803-5519-AEDD-D2BEB56F5AB3	Software Name 43	2023-01-01	2021-11-19
	100	6A380749-276E-7EFA-CBE4-9DF9C387D2CF	Software Name 48	2023-01-01	2022-07-06
	100	C7E7DC04-2C7C-5435-1EFE-46EC19BA5547	Software Name 6	2023-01-01	2022-07-08
	100	BAA39FC2-D910-9719-0A68-CFA8DD582511	Software Name 5	2023-01-01	2022-03-07
	100	BAA99F67-96A6-F013-0B3C-629BEFB4DDFD	Software Name 6	2023-01-01	2022-07-08
	100	76F956F1-3AB7-399F-6B4D-7DA37EC9992E	Software Name 56	2023-01-01	2022-07-09
	100	E840EA1E-7C03-B948-E80B-DCA4901554CE	Software Name 58	2023-01-01	2021-07-02
	100	531267E7-1B84-1929-82C0-6EDDE9C0657E	Software Name 38	2023-01-01	2020-04-21
	100	75D5D829-FF05-1A14-AEBA-06267CB23C12	Software Name 54	2023-01-01	2022-03-11
	100	C6975583-AED4-11CA-D1CF-E42BC29DEA29	Software Name 5	2023-01-01	2022-03-07
	100	59D7B8B1-5B08-7266-C86A-18CD08ECCB22	Software Name 43	2023-01-01	2021-11-19
	100	6F36EEE2-A1DA-6169-694C-B5614C50ADC9	Software Name 50	2023-01-01	2022-04-27

Query 8 – Hardware systems, which have passed their warrantee periods.

Query in English:

Show the hardware systems whose warranties, relative to a target date, have ended. Produce the query results for each of the target dates of "2010-01-01" and "2015-01-01", respectively. (ALL ROWS)

Sample Query Result:

HwSerialNo	HwType	TargetDate	HwsPurchaseDate	WarrDuration	WarEndDate
------------	--------	------------	-----------------	--------------	------------

HwSerialNo	HwType	TargetDate	HwsPurchaseDate	WarrDuration	WarEndDate
0FCDF0CD-7E59-AC7E-926D-E4963ED4D131	nulla.	2015-01-01	2013-07-28	1	2014-07-28
243E528C-01B8-4AB9-23D3-8E614853C3C1	molestiepr3	2015-01-01	2010-11-24	2	2012-11-24
2833D1B3-43B7-A852-71AE-16EDC15EDBCB	ipsumpr1	2015-01-01	2011-12-28	3	2014-12-28

Query in SQL:

[Show the query statements]

```

1 • SELECT SerialNo AS HwSerialNo, Type AS HwType, '2010-01-01' AS TargetDate, PurchaseDate AS HwsPurchaseDate, WarrDuration, DATE_ADD(PurchaseDate, INTERVAL WarrDuration YEAR) AS WarEndDate
2 FROM hardware
3 WHERE DATE_ADD(PurchaseDate, INTERVAL WarrDuration YEAR) <= '2010-01-01';
4
5 • SELECT SerialNo AS HwSerialNo, Type AS HwType, '2015-01-01' AS TargetDate, PurchaseDate AS HwsPurchaseDate, WarrDuration, DATE_ADD(PurchaseDate, INTERVAL WarrDuration YEAR) AS WarEndDate
6 FROM hardware
7 WHERE DATE_ADD(PurchaseDate, INTERVAL WarrDuration YEAR) <= '2015-01-01';
8

```

Result:

[Show the result of the query]

HwSerialNo	HwType	TargetDate	HwsPurchaseDate	WarrDuration	WarEndDate
0FCDF0CD-7E59-AC7E-926D-E4963ED4D131	nulla.	2015-01-01	2013-07-28	1	2014-07-28
243E528C-01B8-4AB9-23D3-8E614853C3C1	molestiepr3	2015-01-01	2010-11-24	2	2012-11-24
2833D1B3-43B7-A852-71AE-16EDC15EDBCB	ipsumpr1	2015-01-01	2011-12-28	3	2014-12-28
4145597A-33A8-109C-62D0-8F18EBE3FE9A	tortor	2015-01-01	2010-08-25	4	2014-08-25
835E503D-EA54-E364-CB4B-715C7895AF3B	trasd	2015-01-01	2011-01-05	1	2012-01-05
841170AB-08AE-61B3-616C-1E51EFD45F4E	server	2015-01-01	2010-12-10	4	2014-12-10
A729782B-91DE-C60B-8607-833D8DEB65DA	silver	2015-01-01	2010-12-27	2	2012-12-27
C439D6C6-069E-F6B5-FFBF-2D7082CC396A	et	2015-01-01	2010-11-20	3	2013-11-20
D283E628-EEBA-72AA-DBEB-54E3425D0AE9	ultricespr2	2015-01-01	2012-12-30	2	2014-12-30
E39E215D-6DD6-7A61-3113-DF883865A1BC	tortor	2015-01-01	2012-01-25	2	2014-01-25

Query 9 – Servers and connected hardware.

Query in English:

Show the hardware systems, which are connected to each server. Group and sort the result by the model numbers of the servers and the model numbers of the connected hardware systems, respectively (List 20 rows)

Sample Query Result:

ServerSerialNo	ServerType	ServerModel	hwSerialNo	hwType	hwModel
B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	server	11	18FDD297-179A-D7CE-DDD3-6554F3F25FF8	ante	10
B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	server	11	17608456-4C36-34D9-DCA1-41FC6154770C	tortor	11
B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	server	11	1218D01D-5680-D5C1-252A-6BABC43CE37A	bronse	12

Query in SQL:

[Show the query statements]

```
1 • select h1.SerialNo as ServerSerialNo, h1.Type as ServerType,
2     h1.Model as ServerModel, h2.SerialNo as hwSerialNo, h2.Type as hwType, h2.Model as hwModel
3     from hardware h1, hardware h2, hwconnectedtohw
4     where h2.SerialNo = hwconnectedtohw.SerialNo_1
5     and h1.SerialNo = hwconnectedtohw.ConnectedToSerialNo_2
6     and h1.Type = "server"
7     order by h1.Model asc, h2.Model asc;
```

Result:

[Show the result of the query]

	ServerSerialNo	ServerType	ServerModel	hwSerialNo	hwType	hwModel
▶	B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	server	11	18FDD297-179A-D7CE-DDD3-6554F3F25FF8	ante	10
	B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	server	11	421DA85C-0BA9-7575-E8B3-D33F1E4547A8	lobortis	11
	B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	server	11	3EAF3ED1-7D4E-C18C-1E9E-AD13D6E5DB59	inceptos	11
	B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	server	11	17608456-4C36-34D9-DCA1-41FC6154770C	tortor	11
	B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	server	11	4145597A-33A8-109C-62D0-8F18EBE3FE9A	tortor	12
	B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	server	11	1218D01D-5680-D5C1-252A-68ABC43CE37A	bronse	12
	B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	server	11	BADA5275-EB9A-F965-BC58-3E72C728F383	tortor	13
	B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	server	11	49D158A8-8A28-0EC8-26BA-A68E279C8D57	malesuada	13
	B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	server	11	A72C6302-82D3-64DA-876B-4A6426A893CB	et	14
	B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	server	11	9F359419-951E-C303-B570-E073A174BAB3	iaculis	14
	B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	server	11	C67D1B09-48A1-39B5-254E-42AA23BB48D8	Ut	15
	B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	server	11	DFE5C66C-9F7C-4E8E-BC2B-D39C833E748D	elementum	15
	B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	server	11	74395D42-0A8D-ED77-CD01-AA729E5B1E6C	ligula	15
	B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	server	11	09654018-524F-2243-4F2A-5FA75C49CFCC	tortor	15
	B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	server	11	2AC93A77-58B3-5BF7-7CC9-DA8DDA256D21	rhoncus	16
	B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	server	11	A07E75DB-4A97-31D2-E964-5B69A07E6D3B	nibh.	16
	B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	server	11	8DE76C42-C471-E120-A779-D4561896BFAD	sagittis.	16
	B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	server	11	D9B97B4A-683E-C874-C603-E6D68E1E3352	vulputate	16
	841170AB-08AE-61B3-616C-1E51EFD45F4E	server	11	8BFC2EDA-B1D8-9938-1ED2-2CD8029DBD22	seri	17
	B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	server	11	8BD969E6-4249-A5EF-5498-9D223CA578D4	Cras	17

Query 10 – Servers and Branches.

Query in English:

For each server, show the branches connected to it. Group and sort the result by the server model number and the branch number, respectively (1ST 20 ROWS)

Sample Query Result:

ServerSerialNo	ServerModel	ServerType	zipcode	BrNumber
841170AB-08AE-61B3-616C-1E51EFD45F4E	11	server	90759	122
841170AB-08AE-61B3-616C-1E51EFD45F4E	11	server	90759	123
841170AB-08AE-61B3-616C-1E51EFD45F4E	11	server	90759	124

Query in SQL:

[Show the query statements]


```

1 • select hardware.SerialNo as ServerSerialNo,
2     hardware.Model as ServerModel,
3     hardware.Type as ServerType,
4     branch.ZipCode as zipcode,
5     branch.BrNumber
6 from hardware, branch
7 where hardware.Type = "server"
8 and hardware.SerialNo = branch.ServerConnectedTo
9 order by hardware.Model asc, branch.BrNumber asc;

```

Result:

[Show the result of the query]

	ServerSerialNo	ServerModel	ServerType	zipcode	BrNumber
▶	841170AB-08AE-61B3-616C-1E51EFD45F4E	11	server	90759	122
	841170AB-08AE-61B3-616C-1E51EFD45F4E	11	server	90759	123
	841170AB-08AE-61B3-616C-1E51EFD45F4E	11	server	90759	124
	B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	11	server	71853	140
	B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	11	server	71853	141
	B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	11	server	71853	142
	B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	11	server	71853	143
	B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	11	server	71853	145
	B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	11	server	71853	146
	B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	11	server	71853	147
	B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	11	server	71853	148
	B62D1AF2-1FB7-9EA3-D1B3-2A6E886401B8	11	server	71853	149
	D65820E6-B4C3-C4C1-38AC-99D69184C3C1	14	server	72206	126
	D65820E6-B4C3-C4C1-38AC-99D69184C3C1	14	server	72206	127
	D65820E6-B4C3-C4C1-38AC-99D69184C3C1	14	server	72206	128
	D65820E6-B4C3-C4C1-38AC-99D69184C3C1	14	server	72206	129
	D65820E6-B4C3-C4C1-38AC-99D69184C3C1	14	server	72206	130
	D65820E6-B4C3-C4C1-38AC-99D69184C3C1	14	server	72206	131
	D65820E6-B4C3-C4C1-38AC-99D69184C3C1	14	server	72206	132
	DDD66138-59CA-4A78-6461-1CE94776DA8D	17	server	76041	133

What to Turn In

1. This document, which includes your responses. Add your name to the file name after the word Fall2023-
2. One exported **self-contained file**, which contains **your team developed** database schema and data. Refer to the following YouTube video, which describes how a MySQL database file can be exported:
[MySQL Workbench Database Export and Import](#)
3. One exported **self-contained file**, which contains your database schema and data, based on the **dataset I gave you**.